Gaurav Deshmukh

San Ramon, California · deshmukhgaurav523@gmail.com · (408)668-5141 · in/gdesh · gdesh.com

SUMMARY

Versatile Software Engineer specializing in scalable backend infrastructure, machine learning pipelines, and AI-driven analytics. Proven expertise architecting secure data ecosystems and deploying advanced ML models to drive automated decision-making.

EXPERIENCE

FPGA Solutions Engineer

Intel Corporation — **August 2022 - August 2025, Santa Clara, CA**

- Engineered comprehensive software tools for hybrid FPGA client/server verification workflows, accelerating validation across 9+ silicon and IP programs.
- Co-developed a novel NPU transactor for next-gen architectures, optimizing chip-level data transactions to increase system efficiency by 170%.
- Deployed Kubernetes-hosted dashboarding systems, providing full-scope build, debug, and performance tracking for 3+ engineering teams.
- Developed Python and SQL automation scripts for log ingestion, eliminating 10 hours of manual validation overhead per week.

Graduate Technical Intern

Intel Corporation — **June 2021 - October 2021, Santa Clara, CA**

- Architected an automated PostgreSQL data pipeline, extracting Intel log telemetry and securely routing 100+ daily records for downstream analytics.
- Designed and deployed an interactive Grafana dashboard to visualize FPGA build, run, and debug processes, reducing manual analysis time by 160%.
- Implemented a Keras deep-learning regression model for predictive analytics, forecasting system behavior to optimize future hardware workflows.

Data Analyst Intern

Siemens EDA — **June 2020 - September 2020, Fremont, CA**

- Engineered predictive analytics and machine learning models utilizing emulation data, delivering optimization insights that reduced customer power usage by 45%.
- Developed a Python-based ETL pipeline to efficiently extract, process, and consume complex C++ data structures.
- Designed and managed SQL databases to structure, merge, and organize full-scope raw emulation data for downstream analysis.

PROJECTS

Data & Systems Engineer

FYM Technologies · September 2025 - Present

- Manage trackside data and electrical systems for 3 race teams across Toyota GT4, Ferrari Challenge/GT3, and Porsche Cup platforms.
- Analyze real-time telemetry utilizing WinTax and VBOX during race sessions, optimizing driver and vehicle performance by 30%.
- Direct ECU integration and troubleshoot complex electrical systems, ensuring 98% reliability for Teradek live video and data streams.
- Develop ML regression models to translate raw race telemetry into actionable track strategy and driver insights.

AI-Driven Yellow Flag Predictor

January 2021 - Present

- Engineered a Keras neural network binary classifier to generate live yellow flag predictions with 75% accuracy.
- Developed a Selenium data pipeline to scrape real-time IMSA timing data and store telemetry in MongoDB.
- Deployed the predictive model for active IMSA events, optimizing race strategy for FYM Technologies partner teams.

EDUCATION

B.S. in Computer Science with Specialization in Artificial Intelligence

Minor in Management · University of California, Irvine · Irvine, CA · 2022 · 3.6

SKILLS

Python, C++, Java, SQL, Swift

Kubernetes, Docker, Grafana, MongoDB, PostgreSQL

TensorFlow, Keras, Scikit-learn, OpenCV

Automotive technology & motorsports, AI in everyday and creative applications, Soccer, Golf, Basketball, Cryptocurrency